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Started	Wednesday, 21 October 2026, 9:32 PM
Completed	Wednesday, 21 October 2026, 9:36 PM
Duration	4 mins 19 secs
Grade	**14.00** out of 15.00 (**93.33**%)

Question **1**

Correct

Mark 1.00 out of 1.00

If a list contains 1000 elements, how many comparisons would a binary search typically make in the worst case?

a. 100
b. 500
c. 10 ✓
d. 1000

Question **2**

Correct

Mark 1.00 out of 1.00

In linear search, how is the element searched?

a. By dividing the list into halves
b. By using a hash function
c. By comparing each element in the list sequentially ✓
d. By sorting the list first

Question **3**

Correct

Mark 1.00 out of 1.00

What is the advantage of binary search over linear search?

a. Binary search has a lower time complexity on large, sorted lists ✓
b. Binary search does not require dividing the list
c. Binary search can find multiple instances of the target element
d. Binary search works on unsorted lists

Question **4**

Correct

Mark 1.00 out of 1.00

Given an array arr = {45,77,89,90,94,99,100} and key = 99; what are the mid values(corresponding array elements) in the first and second levels of recursion?

a. 90 and 99 ✓

b. 90 and 94

c. 89 and 99

d. 89 and 94

Question **5**

Correct

Mark 1.00 out of 1.00

In the context of searching, what is a successful search?

a. When the element is found in the list ✓

b. When the search algorithm finishes

c. When the list contains duplicate elements

d. When the list is sorted

Question **6**

Correct

Mark 1.00 out of 1.00

In binary search, if the target element is less than the middle element, where does the search continue?

a. In the left sublist ✓

b. In the right sublist

c. In the entire list

d. At the beginning of the list

Question **7**

Correct

Mark 1.00 out of 1.00

What happens when the element is found in linear search?

a. The search starts over from the beginning
b. The search backtracks to find duplicate elements
c. The search stops immediately ✓
d. The search continues until the end of the list

Question **8**

Correct

Mark 1.00 out of 1.00

During a binary search, what happens if the target element matches the middle element?

a. The search ends successfully ✓
b. The search continues in the right sublist
c. The search continues in the left sublist
d. The list is sorted

Question **9**

Correct

Mark 1.00 out of 1.00

The average case occurs in the linear search algorithm

a. When the item is not the array at all
b. When the item is somewhere in the middle of the array ✓
c. When the item is the last element in the array
d. Item is the last element in the array or item is not there at all

Question **10**

Correct

Mark 1.00 out of 1.00

What is the key characteristic of binary search?

a. It always starts from the beginning of the list
b. It can be applied only if the list is sorted ✓
c. It works on unsorted lists
d. It compares elements sequentially

Question **11**

Correct

Mark 1.00 out of 1.00

In binary search, what happens if the middle element does not match the target element?

a. The list is sorted
b. The search stops
c. The search continues in the left or right sublist ✓
d. The search continues from the beginning

Question **12**

Correct

Mark 1.00 out of 1.00

Which of the following scenarios is best suited for applying binary search?

a. When the list is sorted ✓
b. When the list is unsorted
c. When the list contains duplicate elements
d. When the list is very small

Question **13**

Correct

Mark 1.00 out of 1.00

_____sort is the simplest sorting algorithm that works by repeatedly swapping the adjacent elements in case they are unordered in n-1 passes.

a. Bubble ✓
b. Selection
c. Complexity
d. Insertion

Question **14**

Correct

Mark 1.00 out of 1.00

What is the best-case time complexity of linear search?

a. O(log n)
b. O(1) ✓
c. O👎
d. O(n log n)

Question **15**

Incorrect

Mark 0.00 out of 1.00

What type of search would be most appropriate for finding an element in a list that is frequently updated?

a. Interpolation search
b. Hash search ✗
c. Binary search
d. Linear search